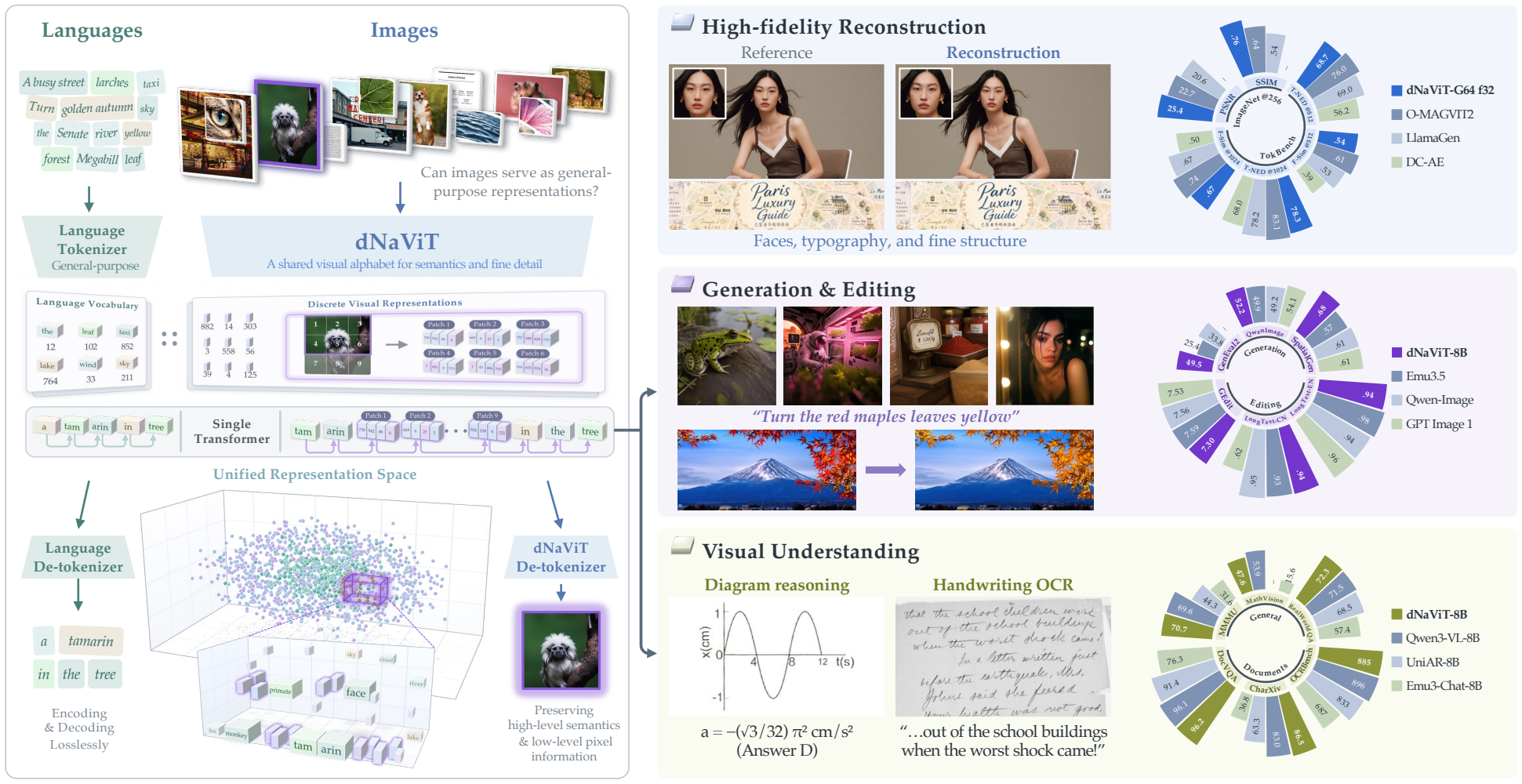

Languages
Images
A busy street
larches
taxi
Turn
golden autumn
sky
the
Senate
river
yellow
forest
Megabill
leaf
Can images serve as general-purpose representations?
Language Tokenizer
General-purpose
dNaViT
A shared visual alphabet for semantics and fine detail
Language Vocabulary
Discrete Visual Representations
Patch 1
Patch 2
Patch 3
Patch 4
Patch 5
Patch 6
Patch 9
a
tam
arin
in
tree
Single Transformer
Unified Representation Space
Language De-tokenizer
dNaViT De-tokenizer
Encoding & Decoding Losslessly
Preserving high-level semantics & low-level pixel information
High-fidelity Reconstruction
Reference
Reconstruction
Faces, typography, and fine structure
ImageNet @256
Tok Bench
SSIM
PSNR
T-NED @512
T-NED @1024
F-Sim @1024
F-Sim @512
dNaViT-G64 f32
O-MAGVIT2
LlamaGen
DC-AE
Generation & Editing
"Turn the red maples leaves yellow"
Generation
Editing
GenEval2
Qwenimage
SpatialGen
GEdit
LongText-EN
LongText-CN
dNaViT-8B
Emu3.5
Qwen-Image
GPT Image 1
Visual Understanding
Diagram reasoning
Handwriting OCR
a = −(√3/32) π² cm/s²
(Answer D)
"…out of the school buildings when the worst shock came!"
General
Documents
MMMU
MathVision
RealWorldQA
DocVQA
CharXiv
OCRBench
dNaViT-8B
Qwen3-VL-8B
UniAR-8B
Emu3-Chat-8B